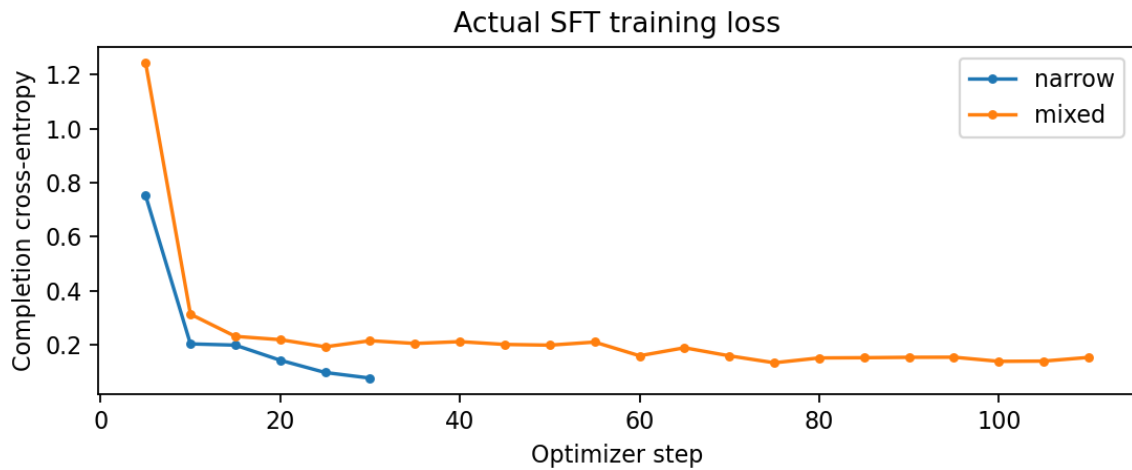


HW5 SFT experiment report

1. Final method

I fine-tuned Qwen2.5-0.5B-Instruct with LoRA on 907 four-choice MMLU questions from machine learning, computer science, college mathematics, statistics, history, geography, nutrition, and elementary mathematics, using at most 120 training questions per subject [1,2]. As in the starter, MMLU test splits supply training data; dev/validation splits are held out, and questions matching either course CSV are removed before training. The final run uses two epochs, LoRA rank 16/alpha 32/dropout 0.05 on all linear layers, AdamW at peak learning rate $1e-4$, weight decay 0.01, cosine decay, 10% warmup, batch 1 with 16-step accumulation, gradient clipping 1, bfloat16 and seed 189 [3]. SFTTrainer minimizes completion-only cross-entropy on boxed answer letters after deterministic choice permutation; base and tuned models use the same constrained letter decoding [4]. Mixing general and specialized examples is motivated by data-composition work [5], but this one-stage experiment does not reproduce DMT.

Model	CS189 public (25)	Specialized holdout	General holdout
base	6/25 (24.0%)	30/75 (40.0%)	73/142 (51.4%)
narrow	12/25 (48.0%)	26/75 (34.7%)	71/142 (50.0%)
mixed	10/25 (40.0%)	28/75 (37.3%)	72/142 (50.7%)



2. Narrow-data attempt

The narrow attempt repeatedly trained only machine-learning questions for 30 optimizer steps at learning rate $3e-4$; its specialized held-out accuracy changed by -5.3 percentage points, while general accuracy changed from 51.4% to 50.0%. Its limitations are consistent with fitting a small, narrow dataset and shifting answer preferences; this comparison changes both data coverage and training budget, so it cannot isolate their causal effects.

Public accuracy is descriptive: the 25-question public set overlaps the hidden CSV. No public or hidden questions were trained on. Kaggle scores are reported on page 2. The held-out MMLU results are local split evaluations, not official benchmark scores.

3-4. Real changes on non-CS189 inputs

Each input uses the same instruction: choose one option and return its letter in a LaTeX box. The examples below are the two shortest observed changes of each type in general held-out subjects. These illustrate individual forgetting and gains, rather than proving a broad loss or gain of an entire skill.

Forgetting 1 | cais/mmlu:elementary_mathematics:validation

Solve for y. $y - 2 + 3y = 10$

A. 2 B. 3 C. 4 D. 5

Base: $\boxed{B}$; final: $\boxed{C}$; correct: B.

The tuned model loses a previously correct linear-equation answer; interference from the limited answer-only training mixture is one possible cause.

Forgetting 2 | cais/mmlu:elementary_mathematics:validation

Solve the equation $29 = x - 17$.

A. 12 B. -12 C. -46 D. 46

Base: $\boxed{D}$; final: $\boxed{C}$; correct: D.

The tuned model chooses the negative of the correct solution, suggesting that the update changed its handling of signs on this input.

Gain 1 | cais/mmlu:elementary_mathematics:validation

If 50% of a number is 20, what is 75% of the number?

A. 8 B. 15 C. 30 D. 45

Base: $\boxed{B}$; final: $\boxed{C}$; correct: C.

Exposure to other elementary-mathematics examples may have improved percentage reasoning on this held-out question.

Gain 2 | cais/mmlu:elementary_mathematics:validation

The expression $-105 + (-14) + 34$ simplifies to which of the following?

A. -57 B. -75 C. -85 D. 143

Base: $\boxed{A}$; final: $\boxed{C}$; correct: C.

The related arithmetic training may have improved this signed-sum answer, although the output alone cannot establish the mechanism.

Kaggle status

Kaggle account shown by the user: Shiyu63. The user-provided submission screenshot shows Public Score 0.40476 and Private Score 0.35294, with status Complete (after deadline). These are scored late-submission results, not confirmation of eligibility for course credit.

The score was transcribed from the original screenshot in the conversation. The screenshot image file still needs to be embedded in this PDF.

Sources and acknowledgement

[1] Qwen model card. [2] cais/mmlu. [3] Hu et al., LoRA. [4] TRL SFTTrainer. [5] Dong et al., SFT data composition. OpenAI Codex assisted with code, debugging and writing; results come from actual model runs.